

# Data Cleaning Report

## E-Commerce Orders Dataset - Raw to Clean

### Data Analytics Assignment 1

#### Objective

The goal of this task was to clean a raw e-commerce orders dataset by systematically identifying missing values, duplicate records, and formatting inconsistencies, and correcting them using standard data-preparation techniques in Excel. The source data (Raw Dataset, 1,200 rows × 14 columns) was audited, cleaned, and documented in a companion workbook (EDA\_Cleaning\_Workbook.xlsx) using live formulas **Methodology** COUNTA, COUNTBLANK, COUNTIF, SUMPRODUCT, TRIM, PROPER, and ROUND rather than manual, one-off fixes, so every check can be re-run automatically if the source data changes.

#### Methodology

The cleaning process followed three stages, each implemented as its own worksheet:

- **Raw Data** the original 1,200 records, unmodified, plus two audit helper columns: a concatenated Row Key (all 14 fields joined into one string) and a Duplicate Flag driven by COUNTIF, used to detect exact duplicate records.
- **Data\_Quality\_Audit** formula-driven checks across four categories: missing/null values, duplicate records, data-format validity (numeric type, date validity, positive values, whitespace), and a summary rollup.
- **Cleaned Data** the final output, rebuilt from Raw Data via formulas that trim whitespace, standardize text casing, validate numeric fields, recompute Total Price, and fill missing values, with a Cleaning Status column documenting the outcome of each row.

#### Step 1 Identifying Missing and Null Values

Every column was checked with COUNTA and COUNTBLANK formulas to count populated versus empty cells. Thirteen of the fourteen columns were completely populated. One column had gaps:

| COLUMN     |       |    | MISSING<br>VALUES | % MISSING | ACTION TAKEN          |
|------------|-------|----|-------------------|-----------|-----------------------|
| COUPONCODE |       |    | 309               | 25.75%    | Filled with 'UNKNOWN' |
| ALL        | OTHER | 13 | 0                 | 0.0%      | No action needed      |
| COLUMNS    |       |    |                   |           |                       |

A blank Coupon Code most likely represents an order placed without a promotional code, so rather than deleting these rows (which would have discarded a quarter of the dataset) or leaving the field blank, missing values were filled with the explicit label 'UNKNOWN'. This keeps every column populated for downstream grouping and aggregation while preserving the distinction between 'no coupon used' and a genuine coupon code.

## Step 2 Removing Duplicates

Duplicate detection was run at two levels: exact-duplicate Order IDs (via COUNTIF) and full-row duplicates, where every one of the 14 fields matches exactly (via a concatenated Row Key checked with COUNTIF).

| Check                            | Result |
|----------------------------------|--------|
| Total rows evaluated             | 1,200  |
| Unique Order Ids                 | 1,200  |
| Duplicate Order IDs found        | 0      |
| Full-row duplicate records found | 0      |

No duplicate records of either kind were found. This step is nonetheless retained as a standing, formula-driven check if the source data were refreshed with a new export, the same audit would automatically flag any duplicates introduced at that point.

## Step 3 Correcting Data Formats

Beyond missing values and duplicates, five categories of format and validity issues were checked using SUMPRODUCT and COUNTIF formulas. All five came back clean, but the corresponding corrections were still applied defensively in Cleaned Data as standard data-preparation practice:

| CHECK                                      | ISSUES FOUND | CORRECTION APPLIED IN CLEANED_DATA                                              |
|--------------------------------------------|--------------|---------------------------------------------------------------------------------|
| NON-NUMERIC QUANTITY / UNIT PRICE          | 0            | Verified numeric type; Unit Price rounded to 2 Decimals                         |
| INVALID OR NON-DATE VALUES IN DATE         | 0            | Standardized display format to YYYY-MM-DD                                       |
| QUANTITY OR UNIT PRICE ≤ 0                 | 0            | No correction required                                                          |
| TOTAL PRICE QUANTITY × UNIT PRICE          | 0            | Total Price recalculated from Quantity × Unit Price on every row as a safeguard |
| LEADING/TRAILING WHITESPACE IN TEXT FIELDS | 0            | All text fields wrapped in TRIM; categorical fields standardized to Title Case  |

Recomputing Total Price from Quantity and Unit Price, rather than trusting the raw value, is a common defensive practice: it costs nothing when the data is already correct (as here) and automatically catches any future data-entry error in a derived field.

## Before /After Summary

|                                   | RAW_DATA | CLEANED_DATA |
|-----------------------------------|----------|--------------|
| ROW COUNT                         | 1,200    | 1,200        |
| MISSING COUPONCODE VALUES         | 309      | 0            |
| DUPLICATE RECORDS                 | 0        | 0            |
| TOTALPRICE CALCULATION MISMATCHES | 0        | 0            |
| TEXT FIELDS WITH STRAY WHITESPACE | 0        | 0            |

## Key Findings

- The raw dataset was already well-formed: no duplicate records, no invalid data types, no negative or zero values, and no calculation errors were present.
- The only genuine data-quality gap was in CouponCode, where 25.75% of orders had no code recorded now explicitly labelled 'UNKNOWN' rather than left blank.
- All text fields (Product, Payment Method, Order Status, Referral Source, etc.) were already consistently cased and free of stray whitespace; TRIM/PROPER formulas were still applied as a standing safeguard.
- Total Price values in the raw data matched Quantity × Unit Price exactly in all 1,200 rows; the cleaned dataset now recalculates this field live rather than trusting a stored value.

## Conclusion

The cleaning workflow converted a mostly-clean but incomplete raw dataset into a fully populated, duplicate-free, consistently formatted dataset ready for analysis. Because every check and correction were implemented as an Excel formula rather than a manual edit, the audit and cleaning steps will automatically re-run and flag new issues if the underlying Raw Dataset is ever refreshed with new records.

Before /After Summary

| OrderID   | Date             | CustomerID | Product | Quantity | UnitPrice | ShippingAddress | PaymentMethod | OrderStatus | TrackingNumber | ItemsInCart | CouponCode | ReferralSource | TotalPrice |
|-----------|------------------|------------|---------|----------|-----------|-----------------|---------------|-------------|----------------|-------------|------------|----------------|------------|
| ORD200789 | 17 08 2023 00:00 | C57276     | Tablet  | 5        | \$691.28  | 183 Main St     | Online        | Delivered   | TRK75899752    | 10          | SAVE10     | Email          | \$3,456.40 |
| ORD201122 | 7 06 2023 00:00  | C38840     | Monitor | 5        | \$678.19  | 766 Main St     | Online        | Returned    | TRK32496970    | 8           | UNKNOWN    | Facebook       | \$3,390.95 |
| ORD200632 | 2 05 2023 00:00  | C67260     | Laptop  | 5        | \$678.16  | 463 Main St     | Gift Card     | Delivered   | TRK38229104    | 7           | WINTER15   | Facebook       | \$3,390.80 |
| ORD200469 | 26 11 2023 00:00 | C13877     | Chair   | 5        | \$676.98  | 893 Main St     | Cash          | Cancelled   | TRK17254691    | 5           | UNKNOWN    | Facebook       | \$3,384.90 |
| ORD200328 | 28 02 2023 00:00 | C18404     | Tablet  | 5        | \$674.04  | 546 Main St     | Online        | Cancelled   | TRK89401624    | 7           | SAVE10     | Google         | \$3,370.20 |
| ORD200107 | 27 03 2023 00:00 | C16775     | Printer | 5        | \$670.75  | 848 Main St     | Gift Card     | Shipped     | TRK34392124    | 8           | FREESHIP   | Instagram      | \$3,353.75 |
| ORD200326 | 1 07 2024 00:00  | C65986     | Laptop  | 5        | \$670.48  | 273 Main St     | Gift Card     | Returned    | TRK98353867    | 5           | SAVE10     | Facebook       | \$3,352.40 |
| ORD201065 | 30 10 2023 00:00 | C47778     | Printer | 5        | \$666.80  | 488 Main St     | Debit Card    | Delivered   | TRK79504329    | 7           | SAVE10     | Referral       | \$3,334.00 |
| ORD201031 | 28 02 2023 00:00 | C59183     | Phone   | 5        | \$664.51  | 136 Main St     | Debit Card    | Pending     | TRK18129706    | 8           | SAVE10     | Email          | \$3,322.55 |
| ORD200463 | 26 05 2023 00:00 | C25276     | Laptop  | 5        | \$662.78  | 214 Main St     | Debit Card    | Shipped     | TRK14011732    | 9           | SAVE10     | Instagram      | \$3,313.90 |
| ORD200361 | 29 06 2024 00:00 | C53464     | Printer | 5        | \$659.85  | 197 Main St     | Credit Card   | Delivered   | TRK32743531    | 9           | SAVE10     | Facebook       | \$3,299.25 |
| ORD200367 | 25 04 2024 00:00 | C13108     | Laptop  | 5        | \$658.77  | 712 Main St     | Gift Card     | Pending     | TRK82284400    | 10          | FREESHIP   | Instagram      | \$3,293.85 |
| ORD200837 | 14 06 2025 00:00 | C88029     | Chair   | 5        | \$655.55  | 282 Main St     | Online        | Delivered   | TRK59701278    | 5           | UNKNOWN    | Email          | \$3,277.75 |
| ORD200527 | 19 08 2024 00:00 | C27202     | Chair   | 5        | \$653.47  | 967 Main St     | Credit Card   | Cancelled   | TRK75406128    | 10          | SAVE10     | Referral       | \$3,267.35 |
| ORD200768 | 4 09 2023 00:00  | C35987     | Tablet  | 5        | \$653.46  | 165 Main St     | Cash          | Cancelled   | TRK28918815    | 9           | WINTER15   | Email          | \$3,267.30 |
| ORD200889 | 27 10 2024 00:00 | C36725     | Monitor | 5        | \$650.72  | 350 Main St     | Credit Card   | Cancelled   | TRK16808790    | 8           | WINTER15   | Email          | \$3,253.60 |
| ORD200540 | 29 01 2024 00:00 | C87281     | Laptop  | 5        | \$648.65  | 852 Main St     | Online        | Pending     | TRK81765057    | 7           | UNKNOWN    | Referral       | \$3,243.25 |
| ORD200802 | 17 02 2024 00:00 | C84331     | Chair   | 5        | \$644.64  | 205 Main St     | Gift Card     | Cancelled   | TRK34440714    | 6           | FREESHIP   | Email          | \$3,223.20 |
| ORD200957 | 12 01 2023 00:00 | C73575     | Monitor | 5        | \$643.89  | 851 Main St     | Cash          | Returned    | TRK23387612    | 8           | WINTER15   | Facebook       | \$3,219.45 |
| ORD200086 | 19 06 2024 00:00 | C88205     | Printer | 5        | \$643.03  | 447 Main St     | Online        | Cancelled   | TRK93504166    | 5           | FREESHIP   | Referral       | \$3,215.15 |
| ORD200221 | 23 02 2024 00:00 | C93473     | Tablet  | 5        | \$639.37  | 552 Main St     | Debit Card    | Pending     | TRK50003989    | 6           | SAVE10     | Google         | \$3,196.85 |
| ORD200967 | 6 05 2025 00:00  | C52879     | Tablet  | 5        | \$638.92  | 502 Main St     | Credit Card   | Returned    | TRK67403284    | 5           | FREESHIP   | Google         | \$3,194.60 |
| ORD200296 | 19 06 2024 00:00 | C48453     | Desk    | 5        | \$638.80  | 434 Main St     | Debit Card    | Cancelled   | TRK52110494    | 10          | WINTER15   | Referral       | \$3,194.00 |
| ORD201079 | 20 06 2024 00:00 | C89979     | Phone   | 5        | \$634.00  | 144 Main St     | Gift Card     | Delivered   | TRK99055909    | 8           | UNKNOWN    | Email          | \$3,170.00 |
| ORD200364 | 24 11 2023 00:00 | C85282     | Phone   | 5        | \$628.74  | 740 Main St     | Cash          | Cancelled   | TRK86767206    | 9           | FREESHIP   | Facebook       | \$3,143.70 |
| ORD200764 | 10 03 2024 00:00 | C35983     | Laptop  | 5        | \$627.43  | 806 Main St     | Credit Card   | Cancelled   | TRK14016163    | 10          | FREESHIP   | Email          | \$3,137.15 |
| ORD200010 | 29 12 2023 00:00 | C43443     | Tablet  | 5        | \$625.97  | 333 Main St     | Credit Card   | Returned    | TRK98859248    | 9           | WINTER15   | Instagram      | \$3,129.85 |
| ORD200820 | 27 05 2023 00:00 | C35377     | Phone   | 5        | \$623.81  | 171 Main St     | Cash          | Returned    | TRK17342157    | 6           | FREESHIP   | Email          | \$3,119.05 |
| ORD200097 | 21 01 2025 00:00 | C15624     | Tablet  | 5        | \$619.52  | 922 Main St     | Debit Card    | Pending     | TRK40621141    | 9           | UNKNOWN    | Email          | \$3,097.60 |
| ORD200241 | 16 05 2023 00:00 | C20817     | Chair   | 5        | \$615.67  | 407 Main St     | Online        | Pending     | TRK44164590    | 6           | SAVE10     | Email          | \$3,078.35 |
| ORD200450 | 15 01 2024 00:00 | C36408     | Monitor | 5        | \$615.10  | 160 Main St     | Gift Card     | Returned    | TRK94023253    | 10          | WINTER15   | Instagram      | \$3,075.50 |
| ORD200876 | 9 06 2024 00:00  | C90363     | Chair   | 5        | \$608.91  | 200 Main St     | Gift Card     | Pending     | TRK21534173    | 6           | SAVE10     | Google         | \$3,044.55 |
| ORD200492 | 25 10 2023 00:00 | C39074     | Laptop  | 5        | \$606.52  | 103 Main St     | Debit Card    | Shipped     | TRK13027036    | 6           | SAVE10     | Instagram      | \$3,032.60 |
| ORD200633 | 10 04 2024 00:00 | C78533     | Laptop  | 5        | \$601.73  | 931 Main St     | Gift Card     | Cancelled   | TRK54711433    | 5           | WINTER15   | Instagram      | \$3,008.60 |